

ZHICHAO “CARTON” ZENG

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UAT keywords: Dark matter | Galaxy dark matter halos | N-body simulations

PROFESSIONAL EXPERIENCE

Texas A&M University

Postdoctoral Associate

– PI: Louis Strigari

College Station, Texas, USA

9/2024-present

EDUCATION

The Ohio State University

Ph.D. in Physics

– Thesis: Self-interacting dark matter subhalos and dwarf satellite galaxies: evolution and diversity

– Advisor: Annika H. G. Peter

Columbus, Ohio, U.S.

8/2018 – 8/2024

The Chinese University of Hong Kong

M.Phil. in Physics

– Thesis: Effects of cosmological neutrinos on the large-scale structure formation

– Advisor: Ming-Chung Chu

Hong Kong

8/2016 – 6/2018

The Chinese University of Hong Kong

B.Sc. in Physics

– Thesis: Influence of massive and degenerate neutrinos on the evolution of the Universe

– Advisor: Ming-Chung Chu

Hong Kong

9/2012 – 6/2016

RESEARCH INTERESTS

Theoretical/computational modelling: dark matter, galactic dynamics, halo–subhalo dynamics, structure formation

Observational probes: dwarf (satellite) galaxies, strong lensing, stellar streams, massive black holes

Methods and tools: N-body simulation (Arepo, Gadget2, Amiga halo finder), semi-analytic/hybrid models

PUBLICATIONS & PRE-PRINTS

Total Publications (15)

Total Citations: 464

Leading Author Publications (6)

Total Citations: 171

- [1] **Z. C. Zeng**, A. Peter, L. Strigari, A. Benson, X. Du, C. Mace, *Feeding the Void: Co-evolution of the SIDM-seeded Black Hole and the dark halo after core-collapse*, 2026, arXiv:2609.22758. *cited by: 0*
- [2] **Z. C. Zeng**, O. Hartl, L. Strigari, A. Peter, X. Du, C. Mace, A. Benson, *Bypassed Core Formation in Milky Way-Mass SIDM Halos: Implications for the Local Group Past-Pericenter Scenario*, 2026, Phys.Rev.D, arXiv:2604.08647. *cited by: 2*
- [3] **Z. C. Zeng**, A. Peter, X. Du, A. Benson, J. Li, et al., *Diversity and universality: evolution of dwarf galaxies with self-interacting dark matter*, 2025, Phys.Rev.D, arXiv:2412.14621. *cited by: 14*
- [4] **Z. C. Zeng**, A. Peter, X. Du, S. Yang, A. Benson, et al., *Till the core collapses: the evolution and properties of self-interacting dark matter subhalos*, 2025, Phys.Rev.D, arXiv:2310.09910. *cited by: 30*
- [5] **Z. C. Zeng**, A. Peter, X. Du, A. Benson, S. Kim, et al., *Core-collapse, evaporation and tidal effects: the life story of a self-interacting dark matter subhalo*, 2022, MNRAS, arXiv:2110.00259. *cited by: 105*
- [6] **Z. C. Zeng**, S. Yeung and M.-C. Chu, *Effects of neutrino mass and asymmetry on cosmological structure formation*, 2019, JCAP, arXiv:1808.00357. *cited by: 20*

Mentored Student Publications (3)**Total Citations: 60**

- [1] C. Mace, B. Dhanasingham, **Z. C. Zeng**, et al., *The Sensitivity of Substructure Lensing to SIDM Core-collapse Model Variation*, 2026, Phys.Rev.D, arXiv:2605.24174. *cited by: 3*
- [2] C. Mace, S. Yang, **Z. C. Zeng**, et al., *Calibrating the SIDM Gravothermal Catastrophe with N-body Simulations*, 2026, Phys.Rev.D, arXiv:2504.13004. *cited by: 14*
- [3] C. Mace, **Z. C. Zeng**, A. Peter, et al., *Convergence Tests of Self-Interacting Dark Matter Simulations*, 2024, Phys.Rev.D, arXiv:2402.01604. *cited by: 43*

Collaborative Publications (6)**Total Citations: 233**

- [1] A. Engelhardt, et al. (inc. **Z. C. Zeng**), *MARVELously Dark: the density profile evolution of dwarf halos in velocity-dependent SIDM*, 2026, JCAP, arXiv:2601.23264. *cited by: 8*
- [2] X. Du, A. Benson, **Z. C. Zeng**, et al., *Tidal evolution of cored and cuspy dark matter halos*, 2024, Phys.Rev.D, arXiv:2403.09597. *cited by: 34*
- [3] S. Yang, et al. (inc. **Z. C. Zeng**), *A quantitative comparison between velocity dependent SIDM cross-sections constrained by the gravothermal and isothermal models*, 2024, MNRAS, arXiv:2305.05067. *cited by: 28*
- [4] S. Yang, X. Du, **Z. C. Zeng**, A. Benson, F. Jiang, E. O. Nadler, A. Peter, *Gravothermal Solutions of SIDM Halos: Mapping from Constant to Velocity-dependent Cross Section*, 2023, ApJ, arXiv:2205.02957. *cited by: 82*
- [5] F. Jiang, et al. (inc. **Z. C. Zeng**), *A semi-analytic study of self-interacting dark-matter haloes with baryons*, 2023, MNRAS, arXiv:2206.12425. *cited by: 63*
- [6] R. Emami, et al. (inc. **Z. C. Zeng**), *Evidence of Neutrino Enhanced Clustering in a Complete Sample of Sloan Survey Clusters, Implying $\sum m_\nu = 0.119 \pm 0.034$ eV*, 2017, arXiv:1711.05210. *cited by: 18*

PRESENTATIONS**Self-interacting dark matter subhalos and dwarf satellite galaxies**

- Poster, Galactic Frontiers II, Dartmouth College June 2025
- Poster, Gravity in the Local Group, CMU June 2025
- Plenary talk, Theoretical Astroparticle and Cosmology Symposium in Texas, Texas A&M Oct 2024
- Texas A&M Astro Seminar Sept 2024
- Plenary talk, Dark Matter 2024, KITP, UCSB Jun 2024
- Ohio State HEP Seminar Mar 2024

Till the core collapses: evolution of SIDM subhalos

- Plenary talk, Galactic Frontiers, CCA Flatiron Institute July 2023
- Seminar (virtual), Nature of Dark Matter, Yale June 2023

Core-collapse, evaporation and tidal effects: the life story of a self-interacting dark matter

- Plenary talk (virtual), N-body-shop workshop, CCA Flatiron Institute June 2022
- Seminar (virtual), The McWilliams Center for Cosmology, CMU Apr 2022
- Poster (virtual), COSMO21 July 2021

MENTORING

Steven Zhu (Undergraduate Student at OSU & UMich)	2025 summer
Odelia Hartl (Graduate Student at TAMU)	2024 – 2026
Chloe Zheng (Graduate Student at OSU)	2024 – Present
Angelica Whisnant (Graduate Student at OSU)	2024 – Present
Charles Mace (Graduate Student at OSU)	2021 – 2026
Jiashen He (Undergraduate Student at OSU)	2019 summer

TEACHING EXPERIENCE

Graduate Teaching Assistant

2018, 2019, 2022

The Ohio State University

Columbus, OH

- Physics courses: Mechanics, Fluids, Waves (Introductory Physics for Engineers and Physicists).
- Astronomy courses: Introduction to Astronomy (General Education) — grading and exam writing; Stellar Evolution — grading.
- Led recitation and laboratory sections, and graded coursework, for undergraduate students.

Graduate Teaching Assistant

2016 – 2018

The Chinese University of Hong Kong

Hong Kong

- Physics courses: Thermodynamics, Computational Physics, General Physics.
- Led recitation and laboratory sections, and graded coursework, for undergraduate students.

AWARDS & HONORS

OSU Presidential Fellowship (Graduate School Dissertation Year Fellowship, \$36k+tuition) 2023 – 2024

Professor Charles K. Kao Research Exchange Scholarship (\$5k) 2013 – 2014

University Full Admission Scholarship (CUHK, \$25k+tuition) 2012 – 2016

PROFESSIONAL SERVICE

Referee: The Astrophysical Journal, Physical Review Letters, European Physical Journal C